

Draft paper — consolidated working document

Working title: *Structural, Linguistic and Veracity Signals of Virality on UK Political Reddit*

How to use this document. All sections are in bullet form for you to expand into prose. Structure follows the department's required layout: Introduction → Related Work → Methodology (Requirements / Design / Implementation) → Results (Experimental results / Evaluation & validation) → Discussion & Conclusions → Future Work → References. Citations use your Zotero keys ([like_this_2021]). [GAP] = find/decide something. [VERIFY] = confirm this against the paper's full text before committing (not assumed). [PLOT N] = figure to include; all figures collected in the Figure List at the end. [DECISION] = needs your/supervisor's call.

1. Introduction

1.1 Context for the project

- Social media is now a primary arena for political discussion → also a vector for large-scale spread of misinformation/malicious content.
- The harm is in the *spread*, not just the existence, of false content — virality turns a fringe claim into a mass belief.
- Reddit specifically: community-structured, vote-based, openly-ranked, archived — a tractable and distinctive setting for studying virality (expanded in 1.4).
- Scope of this project: the drivers of virality on **UK political Reddit**, with veracity (true vs fake) treated as one candidate driver among structural, emotional, political-leaning and topic features.

1.2 Motivation for the work

- Motivating case: the 2024 UK riots. False claims about the Southport attacker's identity spread across platforms via an online-offline feedback loop, amplified by influencers and coordinated/foreign actors; digital falsehoods precipitated offline violence and were recirculated as "proof" ([munk_when_2026]).
- Understanding *what makes political content spread* is therefore of direct public interest.
- Gap that motivates the specific study: the virality/misinformation literature is overwhelmingly US- and Twitter-centric; **to our knowledge no prior work models virality on UK political Reddit**, and veracity is rarely treated as a *feature in a virality model* rather than a standalone detection target ([savatteri_veracity_2026; esteban-bravo_predicting_2024]).

Research questions

- **RQ1:** Which structural, linguistic, emotional, political-leaning, topic and veracity factors predict virality on UK political Reddit?
- **RQ2:** Is **veracity** (true vs fake) a meaningful predictor of virality, and in which direction?
- **RQ3:** Do viral *true* and viral *fake* posts go viral for the *same reasons*? [esteban-bravo_predicting_2024]

1.3 Why Reddit (vs X / Facebook)

- **Community-structured:** content lives in topically-bounded subreddits with their own norms/audiences → virality can be measured *relative to a community*, not a global feed. Underpins the size-normalised virality target.
- **Legible, vote-based mechanism:** Reddit's openly-documented "hot" ranking combines net votes (upvotes – downvotes, log scale) with time decay → engagement reflects explicit community endorsement, not opaque algorithmic amplification (X/Facebook). [GAP: cite Reddit ranking mechanism – Salihefendic's write-up of the open-source algorithm / Cornell INF02040 explainer]
- **Research-tractable:** comparatively accessible, archived, with post-level engagement metadata; Facebook/Twitter impose higher barriers [baumgartner_pushshift_2020].
- **Under-studied:** virality/misinfo literature is Twitter/X-heavy.

1.4 Why UK politics & why 2024

- UK has a distinct media ecosystem, party system and events vs the US-centric literature (Fakeddit, FakeNewsNet, Vosoughi et al. all US).
- 2024 unusually dense for UK political discourse (general election + summer riots) → concentrated signal [antypas_causal_2026; munk_when_2026].
- Window: July–September 2024 captures the 4 July general election, the late-July/Aug riots, and aftermath.
- **Honesty on the riots:** riot misinformation spread principally on X/Facebook/Telegram, *not* Reddit [munk_when_2026] → riots are *societal motivation*; the **general election** is the on-platform driver of discussion volume.
- **[DECISION]** 3-month window is also a compute/scope choice — state plainly (see Limitations in §5/§6).

1.5 Contributions

- A transferable, explainable virality-prediction pipeline, validated US→UK via an explicit cross-demographic comparability analysis.
- The first (to our knowledge) virality analysis of UK political Reddit.

- A SHAP-based account of how veracity interacts with virality in a politically-coherent, text-driven corpus — contrasted with a general US corpus where the same pipeline shows veracity *suppressing* virality.
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2. Related Work

2.1 Virality & popularity prediction on social media

- Barnes et al. — closest methodological precedent: predicts meme popularity on Reddit from content + metadata using tree-based models (best AUC ≈ 0.68); quantifies incremental value of image over text features ([barnes_dank_2021]).
- Dogan et al. — Reddit virality in a cross-lingual, multimodal setting; defines virality by **normalising engagement by community size** and critiques reliance on "static, simple volume-based thresholds with arbitrary cut-offs" ([dogan_early_2026]).
- Chauhan et al. — image/repost cascade dynamics on ideologically diverse subreddits; post- and cascade-level virality prediction ([chauhan_when_2025]).
- Brevity $\rightarrow$ popularity ([GAP: Tsur & Rappoport 2012 not in library – add or drop the specific claim]).

2.2 Veracity, emotion & diffusion

- False news spreads faster/farther/deeper than truth ([noauthor_spread_nodate – Vosoughi, Roy & Aral 2018; fix metadata]).
- Shift from *detecting veracity* to *predicting diffusion*: Savatteri et al. compare the two tasks head-to-head across datasets ([savatteri_veracity_2026]); Esteban-Bravo et al. show content features affect virality *differently* for true vs fake news $\rightarrow$ motivates the subgroup analysis ([esteban-bravo_predicting_2024]).
- Emotion as explanatory variable: negativity retweeted more, incl. UK politicians ([antypas_negativity_2023]); moral emotions moderate misinformation virality (false rumours outspread truth *when* high moral-emotional load) ([solovev_moral_2022]).
- Frameworks: dual emotion (publisher + social/comment emotion) for fake-news detection ([zhang_mining_2021]); emotional signals aid credibility (LSTM — cite for motivation, not architecture) ([giachanou_leveraging_2019]).

2.3 Fake-news detection & datasets

- Fakeddit — large multimodally-labelled Reddit fake-news benchmark ([nakamura_fakeddit_2020]); FakeNewsNet — news + social-context repository ([shu_fakenewsnet_2020]).
- Fine-tuned transformers the current standard for text-based detection ([mouratidis_misinformation_2025 – VERIFY "best practice" claim;])

[devlin_bert_2019](#)]).

- Weak supervision as a route to labels where none exist ([\[ratner_snorkel_2017; shu_early_2021\]](#)) (relevant to earlier UK weak-labelling exploration).

2.4 Political stance & leaning

- Trigger words in UK political discussion → higher engagement + animosity ([\[antypas_causal_2026\]](#)).
- Zero-shot stance extraction from Reddit text ([\[togay_large_2026\]](#)), built on entailment-based zero-shot classification ([\[yin_benchmarking_2019\]](#)) and LLM-based media assessment ([\[carvalho_automated_2026\]](#)).
- Source bias / factual-reporting ratings predictable and operationalisable at scale ([\[baly_predicting_2018; sanchez-cortes_mapping_2024\]](#)).

2.5 Explainable ML for misinformation

- SHAP — additive, game-theoretic attributions; TreeSHAP exact for tree models ([\[lundberg_unified_2017\]](#)).
- Standard explainability tool for fake-news/hate-speech detection ([\[gongane_survey_2024\]](#)), applied on gradient-boosted/RF misinformation models ([\[mouratidis_misinformation_2025; noauthor_pdf_2026 – Hashmi et al. 2024, fix metadata\]](#)).

2.6 Research gap (synthesised)

- US/Twitter-centric literature; virality work rarely models *veracity as a feature*; no prior work on *UK political Reddit* virality → the space this project occupies.
 - **[GAP — suggested additions in the "Additional papers" appendix: FACTOID, UKElectionNarratives, EmoLex/VADER/BERTopic/PR-AUC method cites.]**
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3. Methodology

3.1 Requirements capture & analysis (what the system should do)

- **Primary objective:** given a Reddit post's content + metadata, predict whether it will go viral within its community, and *explain* which features drive that prediction.
- **Functional requirements:**
 - Ingest Reddit data (Fakeddit US; Pushshift UK) and produce a clean, feature-rich post-level dataset.
 - Define a principled, community-normalised virality target that is comparable across subreddits of different sizes.

- Produce a veracity signal for each post (ground-truth for US; model-predicted proxy for UK).
- Train and evaluate virality classifiers with explicit handling of class imbalance.
- Produce per-feature explanations (SHAP) supporting RQ1-RQ3.
- **Non-functional / constraints:**
 - Must run within a limited GPU/compute budget (drives model-size and subsampling choices).
 - US→UK transfer must be *justified*, not assumed — a comparability analysis is a requirement, not an optional extra.
 - No data leakage: virality threshold and any fitted transforms derived from training data only.
- **Success criteria:** BERT veracity classifier competitive with the Fakeddit benchmark (~82%); virality models evaluated on imbalance-appropriate metrics (PR-AUC primary); interpretable SHAP output that answers the RQs.

3.2 Design (how the work is structured)

- **Two-pipeline architecture:**
 - **Pipeline A — Veracity classifier (BERT):** learns true/fake from Fakeddit; used (i) to establish the veracity feature on US data and (ii) to generate predicted veracity labels for UK data.
 - **Pipeline B — Virality predictor (tree ensembles + SHAP):** consumes structural, linguistic, emotional, political-leaning, credibility and veracity features to predict virality and explain it.
- **Cross-demographic comparability (design gate before transfer):** lexical, semantic and engagement-distribution comparisons establish that a semantic veracity model can transfer US→UK.
- **Data sources & corpora:**
 - **US (pipeline development): Fakeddit** ([nakamura_fakeddit_2020]). Multimodal-only subset (human-written title + image) → filters platform noise, preserves benchmark parity. ~564k train / 59k val / 59k test.
 - **UK (target): Pushshift** ([baumgartner_pushshift_2020]) — r/ukpolitics, r/LabourUK, r/tories, r/reformuk, r/uknews (+ r/unitedkingdom present — [DECISION: confirm final list]), Jul-Sep 2024.
 - **[PLOT 1]** Post counts per subreddit (US & UK) / posts-per-month. **[PLOT 2]** Fakeddit label balance + true/fake split.
- **Cross-demographic comparability analysis (notebook 01):**
 - **Lexical:** Jensen-Shannon divergence on unigrams, raw vs entity-blinded (spaCy NER blinding of PERSON/NORP → generic tokens). Raw JSD high (country-specific vocab); blinding reduces it. **[VERIFY on more data — JSD > 0.5 on small sample.]** **[PLOT 3]** JSD raw vs blinded + vocab-intersection.

- *Semantic*: sentence embeddings + UMAP → one shared "cloud" with country-specific corners (Trump/Brexit), not two clusters; domain-classifier cross-val accuracy as overlap measure. [PLOT 4 — KEY] UMAP US vs UK.
- *Engagement shape*: log-score distributions right-skewed/log-normal in both; UK right-shifted with a low-engagement spike; after removing the spike, shapes align. Absolute thresholds differ (US top-20% ≈ 40 ; UK ≈ 154) but target is a *within-corpus percentile* → absolute differences don't affect transferability. [PLOT 5 — KEY] Log-engagement distributions, r/politics vs r/ukpolitics.
- *Conclusion*: lexically distinct, semantically & behaviourally comparable → semantic (BERT) veracity transfer justified.

3.3 Implementation (techniques, problems, solutions)

3.3.1 Pipeline A — Veracity classifier (BERT)

- **Task**: 3-way Fakeddit label (0 true; 1 fake/true-text; 2 fake/false-text) [nakamura_fakeddit_2020].
- **Baseline**: TF-IDF + logistic regression. Best (unweighted): **accuracy 0.77, weighted F1 0.78, macro F1 0.68**. class_weight=balanced improved fake recall but worsened all else (few strong lexical cues for fake diluted by equal weighting).
- **Main model**: (bert-base-uncased), fine-tuned end-to-end [devlin_bert_2019]. Base vs large trades minimal performance for compute [devlin_bert_2019 – VERIFY §5.2]; uncased suits Reddit casing + Fakeddit lowercased titles.
- **Training**: grid over batch {16,32} $\times$ lr {2,3,5}e-5 [devlin_bert_2019 – VERIFY App A.3]; AdamW (β_1 0.9, β_2 0.999, ϵ 1e-6), weight decay 0.01, 10% warmup, dropout 0.1, mixed-precision GradScaler [GAP: Micikevicius et al. 2018]. Best **bs=32, lr=3e-5** (val macro-F1 0.866 @ epoch 2); full retrain → **val macro-F1 0.891**.
- **Problem/solution**: severe class imbalance (fake $\sim 2.5\%$) → report **MCC** + per-class metrics, not just accuracy [mouratidis_misinformation_2025 – VERIFY MCC-for-imbalance].

3.3.2 Pipeline B — Virality target construction (notebook 05)

- Raw engagement = (score + num_comments) [chauhan_when_2025 – VERIFY the formula is actually in the text; else attribute to first principles].
- Size-normalisation (no subscriber counts → subreddit-median proxy, **own operationalisation**): ($\ln(\text{raw}+1)$), then within-subreddit z-score; clip negative raw to 0. Rationale: raw engagement power-law → log $\approx$ normal → z-scoring valid; within-subreddit z = "how many SDs above typical for this community?". Principle of community-size normalisation follows [dogan_early_2026; barnes_dank_2021 – VERIFY Barnes's exact method]; the log-z-within-subreddit formula is our own.
- **Threshold**: top-10% (90th percentile) of normalised engagement, **training set only**, applied unchanged to val/test (no leakage). $z \approx 1.36$ (US) / 1.55 (UK).

- **Problem/solution — k-means rejected:** distribution unimodal → k=2 degenerates to a near-median split (would call ~55 raw engagement "viral"). Reverted to percentile thresholding, echoing Dogan's critique of arbitrary cut-offs while noting their *dynamic* hybrid-score approach is unavailable on static snapshots ([dogan_early_2026]). **[DECISION: top-5% vs top-10% — sensitivity check.]** **[PLOT 8 — KEY]** raw power-law → normalised z + threshold.

3.3.3 Pipeline B — Feature engineering

- **Structural/temporal** ([barnes_dank_2021 – VERIFY exact feature set]): title word count; hour-of-day (4h bins), day-of-week, is_weekend (US Central / GMT); has_image (native Fakeddit; engineered from URL for UK); caps ratio; punctuation densities (?,!, repeated) normalised by length (UK; stylometric, kept separate from VADER for explainability ([GAP: stylometry→engagement cite; or argue from first principles – antypas_negativity_2023 / solovev_moral_2022 cover broader emotion→virality])).
- **Domain credibility:** MBFC bias + factual-reporting via Idiap dataset ([sanchez-cortes_mapping_2024; baly_predicting_2018]), label-encoded (ordinal). ~92% sparsity (native/unrated) — limitation.
- **Psycholinguistic:** NRC EmoLex 8 emotions ([GAP: Mohammad & Turney 2013 – add]) + VADER compound ([GAP: Hutto & Gilbert 2014 – add]). Implements **only** the publisher-emotion (lexicon + sentiment) part of dual-emotion ([zhang_mining_2021]) — no comment text → no social emotion (limitation). Motivation: emotion aids credibility ([giachanou_leveraging_2019]); false content more emotionally charged ([noauthor_spread_nodate – Vosoughi; solovev_moral_2022]). EmoLex word-level/context-blind (limitation).
- **Political leaning:** US — keyword density ratio (BART-MNLI dropped: US left/right unreliable + compute prohibitive on 564k). UK — trigger words + left/right keyword ratio ([antypas_causal_2026 – VERIFY the five trigger words are theirs]); zero-shot BART-MNLI reserved for UK if compute allows ([togay_large_2026; yin_benchmarking_2019; carvalho_automated_2026]). **[Planned, not done.]**
- **Veracity:** US — ground-truth binary (is_fake) ([nakamura_fakeddit_2020]). UK — **BERT-predicted** (p_fake)/(p_true) from Pipeline A (no UK ground truth), *noisy cross-domain proxy* (limitation).
- **Engineered interaction:** (p_fake × is_low_credibility_domain) (own, to make interaction SHAP-readable).
- **[PLOT 9]** Emotion by veracity class (US) + Mann-Whitney effect sizes. **[PLOT 10]** Feature sparsity / Spearman correlation heatmap.

3.3.4 Pipeline B — Models, evaluation & explainability

- **Baseline:** Balanced Random Forest (imblearn) ([barnes_dank_2021 – VERIFY RF usage]). **Main:** XGBoost with (scale_pos_weight) ([chen_xgboost_2016;

dogan_early_2026 – VERIFY XGBoost]. XGBoost over Barnes's gradient boosting for native imbalance handling, L1/L2 regularisation, speed ([chen_xgboost_2016]).

- 5-fold CV, GridSearchCV; grid search on stratified 50k subsample (kernel/memory limits), then full retrain on best params.
 - Metrics: **PR-AUC (primary), ROC-AUC, F1 @ optimal threshold, confusion matrix.** Accuracy rejected (all-0 dummy $\approx 90\text{--}91.5\%$) ([GAP: Saito & Rehmsmeier 2015; or lean on dogan_early_2026 / savatteri_veracity_2026]).
 - Explainability: **TreeSHAP** ([lundberg_unified_2017]); global beeswarm + dependence + 4-way subgroup (viral/non-viral $\times$ fake/true) ([gongane_survey_2024]).
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4. Results

4.1 Experimental results — Pipeline A (veracity classifier)

- **Test: accuracy 0.895, macro-F1 0.890, MCC 0.793.** Per-class F1: 0.873 / 0.886 / 0.910. Meets/exceeds the $\sim 82\%$ Fakeddit benchmark ([nakamura_fakeddit_2020]).
- **[PLOT 6]** BERT test confusion matrix (3×3 , normalised). **[PLOT 7]** training curves (appendix).

4.2 Experimental results — Pipeline B, US / Fakeddit (baseline)

- **Weak model performance** — RF PR-AUC 0.149 / ROC-AUC 0.616; XGBoost PR-AUC 0.148 / ROC-AUC 0.609. Interpreted as a *dataset* property (Fakeddit image-driven; text+metadata carry little engagement signal), not model failure. **[PLOT 11]** US PR + ROC (RF vs XGB).
- **SHAP global:** time_of_day dominant ($|\text{SHAP}|$ 0.325); is_fake **2nd (0.127)**; title_word_count 3rd (0.076, brevity). Emotion weak; MBFC negligible (sparsity); has_image / is_weekend ≈ 0 (redundant/bug $\rightarrow$ drop). **[PLOT 12 — KEY]** US SHAP beeswarm.
- **Key direction (RQ2, US):** fake content *suppresses* virality (mean SHAP -0.168 fake vs $+0.091$ true) — on Fakeddit, true content more viral. Coherent given fake class $\sim 85\%$ image-manipulation/satire. **[PLOT 13]** is_fake SHAP dependence.
- **Subgroup (RQ3, US):** viral-fake and viral-true do **not** go viral for meaningfully different reasons — both driven by time and length; veracity a consistent suppressor, not a selective amplifier. **[PLOT 14]** subgroup SHAP.

4.3 Experimental results — Pipeline B, UK political Reddit (preliminary)

- Threshold $z \approx 1.55$ (≈ 231 raw); 10.5% viral.
- **Domain:** external links far more viral than native (15% vs 3%, +11.3pp); polarised (right/right-centre) domains out-engage neutral; mixed-factuality out-engages high-factuality. **[PLOT 15]** virality rate by MBFC bias $\times$ factuality.

- **Emotion (viral vs non-viral, univariate):** viral posts significantly **more negative** VADER (-0.150 vs -0.037); anger +32%, fear +26%, disgust +31%, sadness +30% — consistent with `[antypas_negativity_2023; solovev_moral_2022]`. [PLOT 16] emotion viral vs non-viral (UK).
- **Model (structural features only; p_fake/topics not yet added):** XGBoost macro-F1 ≈ 0.50 – 0.53 at tuned threshold. [PLOT 17] UK PR/ROC.
- **SHAP (structural only):** `[is_external_link]` most important (news vs discussion); `[word_count]` (brevity); `[caps_ratio]` suppresses (likely spam/clickbait removed/downvoted); `[vader_compound]` *positive* $\rightarrow$ viral (counterintuitive); `[q_density]` suppresses; emotion near-zero; MBFC minimal (sparsity). [PLOT 18 — KEY] UK SHAP beeswarm (after p_fake). [PLOT 19 — KEY] US vs UK feature-importance comparison.

4.4 Evaluation & validation

- **Metric justification:** PR-AUC primary given $\sim 10\%$ positive class; accuracy misleading (dummy $\approx 90\%$). ROC-AUC + F1@threshold + confusion matrix as supporting views.
 - **No-leakage validation:** threshold + transforms fit on train only, applied to val/test.
 - **Transfer validation:** cross-demographic analysis (3.2) as the explicit check that US $\rightarrow$ UK transfer is legitimate.
 - **[PLANNED — evaluation gaps to close, see §6]:** significance testing (Mann-Whitney + effect sizes), model-comparison CIs (bootstrap / DeLong), threshold sensitivity (5% vs 10%), and — critically — **validation of the transferred veracity classifier on a hand-labelled UK gold set** (currently unvalidated on UK).
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5. Discussion & Conclusions

5.1 Critical analysis of findings

- **Veracity matters, even where the model is weak:** on US Fakeddit, `[is_fake]` is the 2nd most important feature despite poor overall model performance — content veracity carries signal for virality.
- **But direction is dataset-dependent:** veracity *suppresses* virality on Fakeddit — almost certainly because the fake class is $\sim 85\%$ image-manipulation/satire (a general meme corpus), not political misinformation. This is the central caveat of the US result and the reason the UK analysis exists.
- **UK structural signal is intuitive and coherent:** external news links, brevity, and (spam-suppressed) caps behave as theory predicts; the corpus behaves like a news-sharing community.
- **An unresolved tension:** univariate emotion analysis (viral = more negative/high-arousal) conflicts with multivariate SHAP (positive VADER $\rightarrow$ viral). Likely

confounding — external news links are both positive-sentiment and viral. Must be resolved before strong emotion claims (partial-dependence check, §6).

5.2 Honest appraisal (limitations)

Data & scope

- Only 3 months of UK data → subreddit-level z-normalisation unstable for small communities (e.g. GreenPartyUK). Expected to smooth with more data.
- UK subreddit selection is a judgement call; limited external validity.
- Pushshift captures posts near creation → late-accruing engagement may be undercounted ([baumgartner_pushshift_2020]).
- The 2024 window is partly a compute/scope decision.

Veracity label (most significant)

- UK (p_fake) is an **unvalidated cross-domain proxy** (US image-heavy Fakeddit → UK text-only politics). Large domain shift; treat all (p_fake) results with caution until validated on a UK gold set.
- Fakeddit's fake class composition makes the US "fake suppresses virality" result a weak comparator for political misinformation.

Features

- Only publisher emotion implemented; social-emotion half of dual-emotion ([zhang_mining_2021]) unavailable (no comment text).
- EmoLex word-level/context-blind.
- MBFC features ~92% sparse → weak signal.
- Stylometric (punctuation/caps) features lack a direct citation.
- Political-leaning & topic features **not yet in the UK model** → current UK results are structural-only and provisional.

Modelling & interpretability

- Weak US model performance (PR-AUC ≈ 0.15) → low baseline ceiling.
- Grid search on subsamples → possibly sub-optimal hyperparameters.
- SHAP explains the *model*, not ground truth; weak model $\Rightarrow$ read attributions as "what the model used".

5.3 Conclusions

- A transferable, explainable virality pipeline has been built and validated US→UK at the comparability level; the veracity classifier is strong (test macro-F1 0.89).
- Preliminary evidence: veracity is a non-trivial virality signal, but its direction and rank are corpus-dependent — motivating the UK-specific analysis that is the project's core

contribution.

- **[Complete after UK model finished:]** state the final answer to RQ2/RQ3 for UK political Reddit once p_fake + topics are in the model.
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6. Future Work

6.1 The fake-news label — central open decision

- **(A) Keep Fakeddit-trained BERT, apply to UK as a noisy proxy (current plan).**
Cheapest; classifier already strong. Needs honest caveats + UK gold-set validation.
Recommended default.
- **(B) Validate/supplement with a UK-appropriate source:** FACTOID (Reddit political, credibility + bias labels; $\text{[GAP: Sakketou et al. 2022 - add]}$) and/or UKElectionNarratives ([GAP: add]) → build a UK gold test set / weak labels. Medium cost, big validity gain.
- **(C) Full swap of Fakeddit for a political dataset.** Highest cost (re-train Pipeline A, redo transfer analysis, lose benchmark parity). Only if time allows.
- **[DECISION — supervisor]** My steer: A + B.

6.2 Feature completion (UK)

- Add p_fake / p_true to the UK matrix (**the core cross-pipeline feature — currently missing**).
- BERTopic on UK train, applied to val/test ($\text{[GAP: Grootendorst 2022 - add]}$); topic-segmented SHAP (do viral rates / the p_fake -virality relationship differ by topic?).
[PLOT 21] virality & p_fake -SHAP by topic cluster.
- Finalise political-leaning method (keyword ratio vs Antypas triggers vs BART-MNLI).

6.3 Statistical significance testing (plan)

- Group differences (viral vs non-viral; fake vs true): Mann-Whitney U + **rank-biserial r** effect sizes (report effect size, not just p, given large n).
- Multicollinearity: VIF / Spearman heatmap before XGBoost (also helps the emotion tension).
- Model comparison: bootstrap CIs on PR-AUC/ROC-AUC (RF vs XGBoost); DeLong or bootstrap test for significance.
- Threshold sensitivity: re-run at 5% and 10%; check SHAP-ranking stability.
- Core RQ (US vs UK veracity effect): compare p_fake / is_fake SHAP rank + direction across corpora. **[DECISION: descriptive SHAP-rank comparison sufficient, or something inferential expected?]**

6.4 Interactive UI (stretch deliverable)

- Concept: paste a Reddit post/URL → extract features → predict virality probability → SHAP explanation of top contributors + short natural-language report.
- Value: demonstrates the pipeline end-to-end; makes explainability tangible.
- Feasibility: model + SHAP already exist; needs a feature-extraction wrapper + simple front-end (Streamlit/Gradio); (p_fake) needs the BERT checkpoint served.
- [DECISION: dissertation scope or "future work" mention only? Must not displace core analysis + gold-set validation.]

6.5 Robustness & extension

- Hand-label a UK gold test set to validate the transferred veracity classifier (**biggest current validity threat**).
 - Load more months of UK data to stabilise small-subreddit normalisation.
 - Extend to comment-level social emotion if comment text can be obtained (completes the dual-emotion framework).
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7. References

(Generate from Zotero. Fix the metadata/year issues below first.)

Zotero housekeeping (fix before generating references):

- (noauthor_spread_nodate) → Vosoughi, Roy & Aral (2018), *Science* — add author/year.
 - (noauthor_fake_nodate) → 2016-election Twitter *Science* paper (likely Grinberg et al. 2019) — identify/fix.
 - (noauthor_uk_nodate) → UK GE 2024 survey — add author/publisher.
 - (noauthor_pdf_2026) → Hashmi et al. (2024) FastText/Explainable-AI paper — fix.
 - Many (date = 2026) are date-added, not publication year: Devlin 2019, Chen & Guestrin 2016, Barnes 2021, Zhang 2021, Lundberg 2017, Giachanou 2019, Yin 2019, Gongane 2024, Solovev 2022, Antypas (negativity) 2023, Nakamura 2020, etc. **Correct before citing.**
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Appendix A — Additional papers to add (credible, well-cited)

Tool/method cites you currently lack (must add — you use these):

- Mohammad & Turney (2013) — NRC EmoLex.
- Hutto & Gilbert (2014) — VADER.

- **Micikevicius et al. (2018)** — Mixed Precision Training (justifies GradScaler).
- **Grootendorst (2022)** — BERTopic (planned use).
- **Saito & Rehmsmeier (2015)** — PR-AUC on imbalance (anchors metric choice).
- **Tsur & Rappoport (2012)** — brevity/popularity (get source or drop claim).

For the fake-news-label decision / UK domain:

- **Sakketou et al. (2022), FACTOID** — contemporary Reddit political misinfo dataset (credibility + bias labels).
- **UKElectionNarratives (2025)** — UK election misinformation narratives (gold-set seed).

Optional strengtheners:

- Reddit-ranking-mechanism source (Salihefendic / Cornell INFO2040) — for the "why Reddit" claim.
- **Pennycook & Rand** — why people share misinformation (psychology anchor for the intro).
- **Lundberg et al. (2020, Nat. Mach. Intell.)** — TreeSHAP journal version (stronger exactness cite).

Appendix B — Consolidated figure list

KEY FIGURES (carry the argument — prioritise):

- **PLOT 4** — UMAP US vs UK (transfer justification).
- **PLOT 5** — Log-engagement distributions (behavioural comparability).
- **PLOT 8** — Virality target construction (normalisation + threshold).
- **PLOT 12** — US SHAP beeswarm (is_fake 2nd).
- **PLOT 18** — UK SHAP beeswarm (where veracity ranks in UK politics).
- **PLOT 19** — US vs UK feature-importance comparison (central contribution).

#	Section	Figure	Status
1	3.2	Post counts per subreddit / over time	data ready
2	3.2	Fakeddit label balance + true/fake split	data ready
3	3.2	JSD raw vs blinded + vocab intersection	data ready (small sample)
4	3.2	UMAP US vs UK (KEY)	data ready
5	3.2	Log-engagement distributions (KEY)	data ready
6	4.1	BERT test confusion matrix	data ready
7	4.1	BERT training curves (appendix)	data ready
8	3.3.2	Virality target construction (KEY)	data ready
9	3.3.3	Emotion by veracity class (US) + effect sizes	data ready
10	3.3.3	Feature sparsity / Spearman heatmap	to generate
11	4.2	US PR + ROC (RF vs XGB)	data ready
12	4.2	US SHAP beeswarm (KEY)	data ready
13	4.2	<code>is_fake</code> SHAP dependence (US)	data ready
14	4.2	Subgroup SHAP comparison (US)	data ready
15	4.3	Virality rate by domain credibility (UK)	data ready
16	4.3	Emotion viral vs non-viral (UK)	data ready
17	4.3	UK PR + ROC	data ready (update after p_fake)
18	4.3	UK SHAP beeswarm (KEY)	after p_fake + topics
19	4.3	US vs UK feature-importance comparison (KEY)	after UK model complete
20	5.1	VADER partial-dependence (resolve emotion tension)	to generate
21	6.2	Virality & p_fake-SHAP by BERTopic cluster	after topics

Presentation notes: consistent colour scheme for true/fake and viral/non-viral across all figures; label SHAP axes with direction ("→ more viral"); training curves + JSD tables in an appendix to keep the main text on the six key figures.